

The fluctuation closed form:

$$S_\lambda(a) = 2^{1-\lambda}\beta^{-\lambda}\Gamma(2\lambda)e^{\beta a^2/8}D_{-2\lambda}(-a\sqrt{\beta/2})$$

Claude, with Daniel Murfet

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Abstract

Unit 7 of the grammar-paper §4 autoformalisation, and the payoff of the option-B parabolic-cylinder detour: the corollary `cor:fluctuation_closed_form`, which had been the arc’s standing blocker. The change of variables $t = u^2/(2\beta)$ carries Watanabe’s fluctuation function onto the $D_{-2\lambda}$ integral kernel; the Γ and Gaussian prefactors then cancel. Proved via an intermediate Γ -free “kernel” lemma, so that all substitution bookkeeping stays disentangled from the special-function definition. Zero `sorry/axiom`.

1 Setting

Unit 6 built the negative-order parabolic cylinder function `parCylNeg` = $D_{-\nu}(x) = e^{-x^2/4}/\Gamma(\nu) \int_0^\infty u^{\nu-1} e^{-u^2/2-xu} du$ and its integrability. This unit is the reason it was built: the paper’s closed form for S_λ . With it, the entire analytic core of §4 that does not require an operator/commutator framework is formalised — the ladder, the recurrence, the ODE, and now the parabolic-cylinder identification.

2 The things formalised

```
theorem fluctuation_eq_kernel ( lam a : ) (h : 0 < ) :
  fluctuation lam a
    = (2 ^ (1 - lam) *  ^ (-lam))
      * u in Set.Ioi 0,
        u ^ (2*lam - 1) * Real.exp (-u^2/2 + a * Real.sqrt ( /2) * u)

theorem fluctuation_closed_form ( lam a : ) (h : 0 < ) (hlam : 0 < lam) :
  fluctuation lam a
    = 2 ^ (1 - lam) *  ^ (-lam) * Real.Gamma (2*lam) * Real.exp ( *a^2/8)
      * parCylNeg (2*lam) (-a * Real.sqrt ( /2))
```

3 Proof strategy

The substitution $t = u^2/(2\beta)$ is done in two library steps rather than one square-root change of variables, following GPT-6 Astra’s steer. First scale $t \mapsto r/(2\beta)$ with `integral_comp_mul_left_Ioi` (Jacobian $(2\beta)^{-1}$); then apply `integral_comp_rpow_Ioi_of_pos backwards` at $p = 2$ (Jacobian $2u$). Their product, $(2\beta)^{-1} \cdot 2u = u/\beta$, is exactly dt/du . A single pointwise identity on $(0, \infty)$ then matches the transported integrand $2u \cdot F(u^2/(2\beta)) = 2(2\beta)^{-(\lambda-1)} u^{2\lambda-1} e^{-u^2/2+a\sqrt{\beta/2}u}$, and

`integral_const_mul` pulls the constant out. Collecting the two Jacobians and the $(2\beta)^{-(\lambda-1)}$ against the target prefactor is the scalar identity $(2\beta)^{-1} \cdot 2(2\beta)^{-(\lambda-1)} = 2^{1-\lambda}\beta^{-\lambda}$ (`hconstant`). This is `fluctuation_eq_kernel`, which never mentions Γ .

The corollary then unfolds `parCylNeg`($2\lambda, -a\sqrt{\beta/2}$): its kernel integrand $e^{-u^2/2 - (-a\sqrt{\beta/2})u}$ is (after one `ring` rewrite) the kernel of `fluctuation_eq_kernel`, its Gaussian prefactor is $e^{-\beta a^2/8}$ (since $(-a\sqrt{\beta/2})^2 = \beta a^2/2$), and its $1/\Gamma(2\lambda)$ cancels the $\Gamma(2\lambda)$ in the target. `field_simp` closes it with $\Gamma(2\lambda) \neq 0$ and $e^{\beta a^2/8} \neq 0$.

4 Roadblocks and resolutions

The route was Astra-planned, so the friction was purely Lean mechanics:

- The two change-of-variables lemmas live in `namespace MeasureTheory` inside a plain section `IoiChangeVariables` (grep for `namespace` finds only the enclosing one); with `open MeasureTheory` the unqualified names resolve.
- The pointwise power identity splits into three separately-stated `haves` (`hpow`, `hexp`, `hxx`): `ring` cannot see through `rpow`, so each exponent identity $(2(\lambda-1) = 2\lambda-2, 1+(2\lambda-2) = 2\lambda-1)$ is normalised by an explicit `show ... by ring` before `Real.rpow_mul/rpow_add`.
- Two square-root facts are proved by *avoiding* $(\sqrt{\cdot})^2$ under `field_simp`: $\sqrt{2\beta}\sqrt{\beta/2} = \beta$ via `Real.sqrt_mul+sqrt_sq`, and then $\beta/\sqrt{2\beta} = \sqrt{\beta/2}$; substituting the latter backwards keeps the exponent-phase identity rational in $\sqrt{2\beta}$ (never squared), so `field_simp`; `ring` closes it.
- `setIntegral_congr_fun` hands its pointwise goal as applied `lambdas` — `beta_reduce` before the `rw`.
- Several `field_simp` calls closed their goals outright; the trailing `rings` then errored “no goals” and were deleted (a recurring repo pattern).

5 What was Mathlib, what was new

The change-of-variables engine is entirely Mathlib (`integral_comp_mul_left_Ioi`, `integral_comp_rpow_Ioi_of_pos`, `integral_const_mul`, the `rpow/sqrt` API). What is new is the pairing of the two Jacobians to realise the square-root substitution without a square-root change-of-variables lemma, the Γ -free intermediate factorisation, and the prefactor cancellation that identifies S_λ with a named special function Mathlib does not have.

6 Lessons

A square-root change of variables is often cheaper as *scale-then-square* than as one direct substitution: the two off-the-shelf Jacobians multiply to the one you want, and each half is a lemma Mathlib already proves unconditionally. And factoring the special-function definition out of the substitution (prove the bare kernel integral first, unfold the definition only at the end) keeps Γ out of the change-of-variables algebra entirely — the cancellation becomes a two-line `field_simp`.

7 Follow-ups

With the closed form landed, the D_ν arc continues to `eq:Dn_hermite` (the integer-order D_n against Mathlib's Hermite polynomials) and `lem:qho` (the quantum harmonic oscillator identification). The remaining non-special-function §4 items — `prop:fluctuation_weber` (ODE→Weber transform) and the commutator $[b, b^\dagger] = 1$ — are independent of D_ν .